SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No: 1

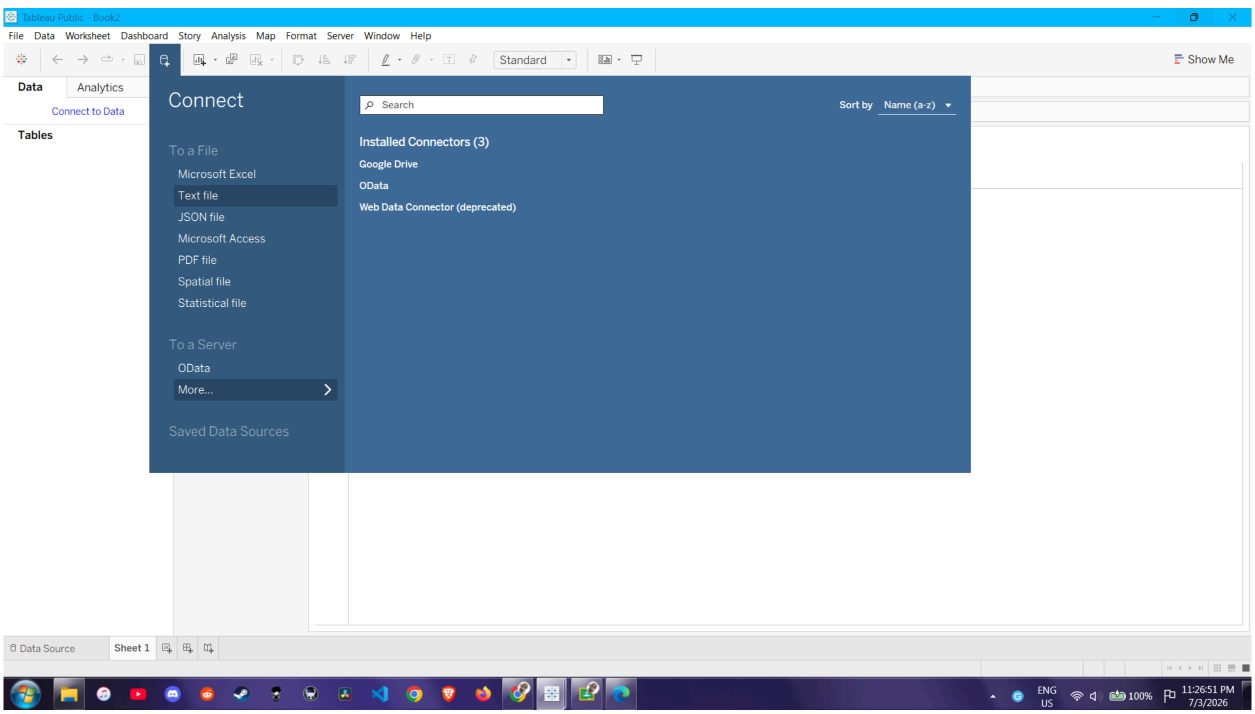
Aim:

- Connect to a CSV dataset and explore the Tableau interface
- Identify and differentiate Measures and Dimensions
- Convert fields between discrete and continuous and observe changes
- Create and customize a basic bar chart
- Build a line chart for time-series analysis
- Create a geographic map visualization
- Combine multiple charts into a simple interactive dashboard

Steps:

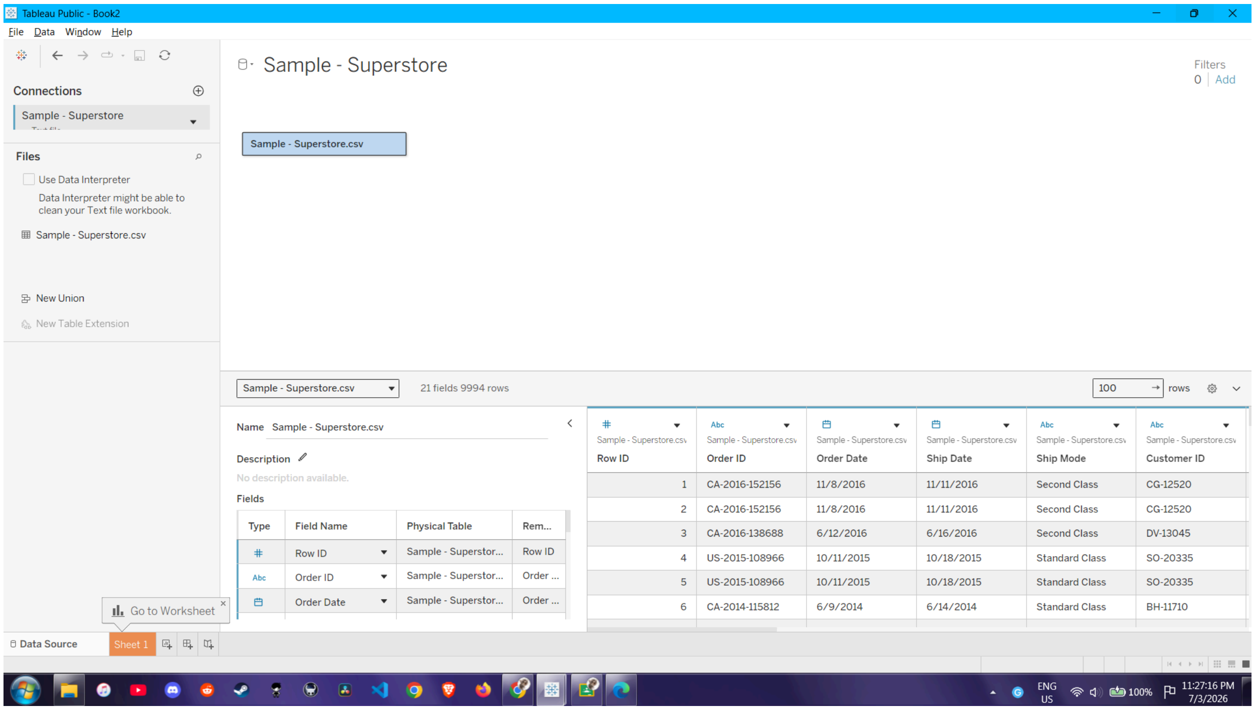
1. Connect to a CSV dataset and explore the Tableau interface

Launch Tableau, connect to your Superstore CSV file via the "Text File" option , preview your data , and click "Sheet 1" to enter your workspace.



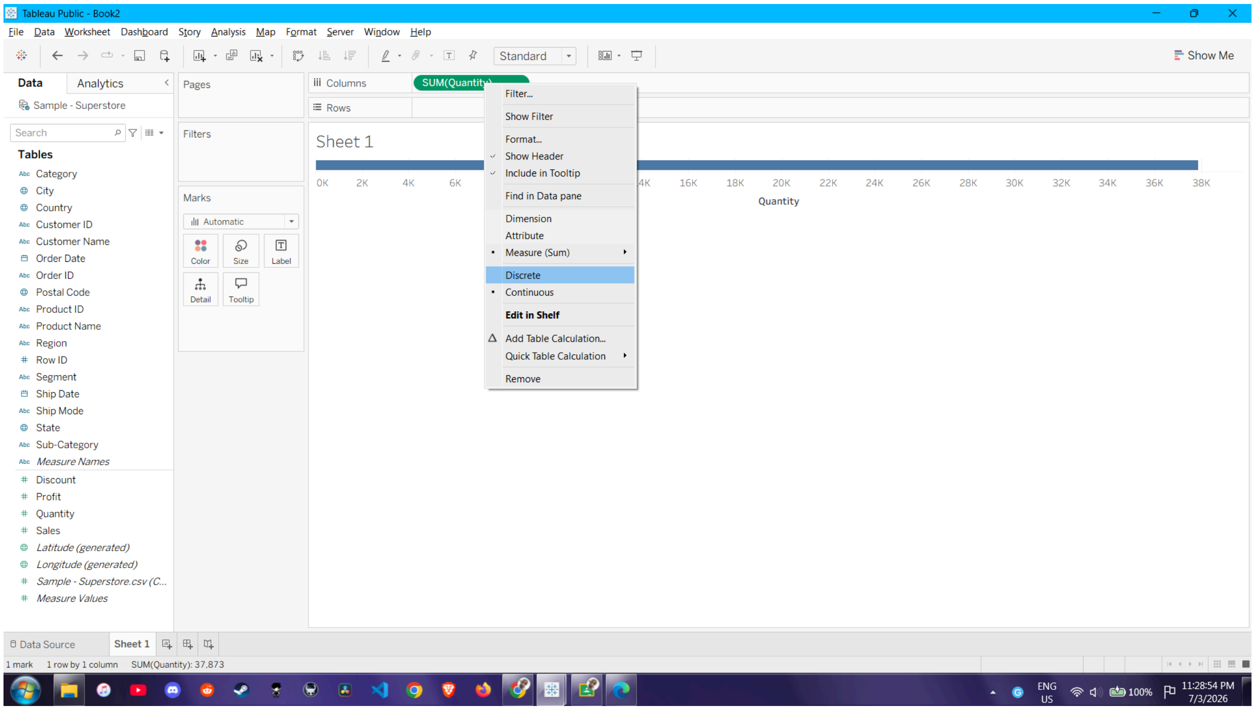
2. Identify and differentiate Measures and Dimensions

In the Data Pane, locate your categorical Dimensions (blue icons, like Category) and your numerical Measures (green icons, like Sales).



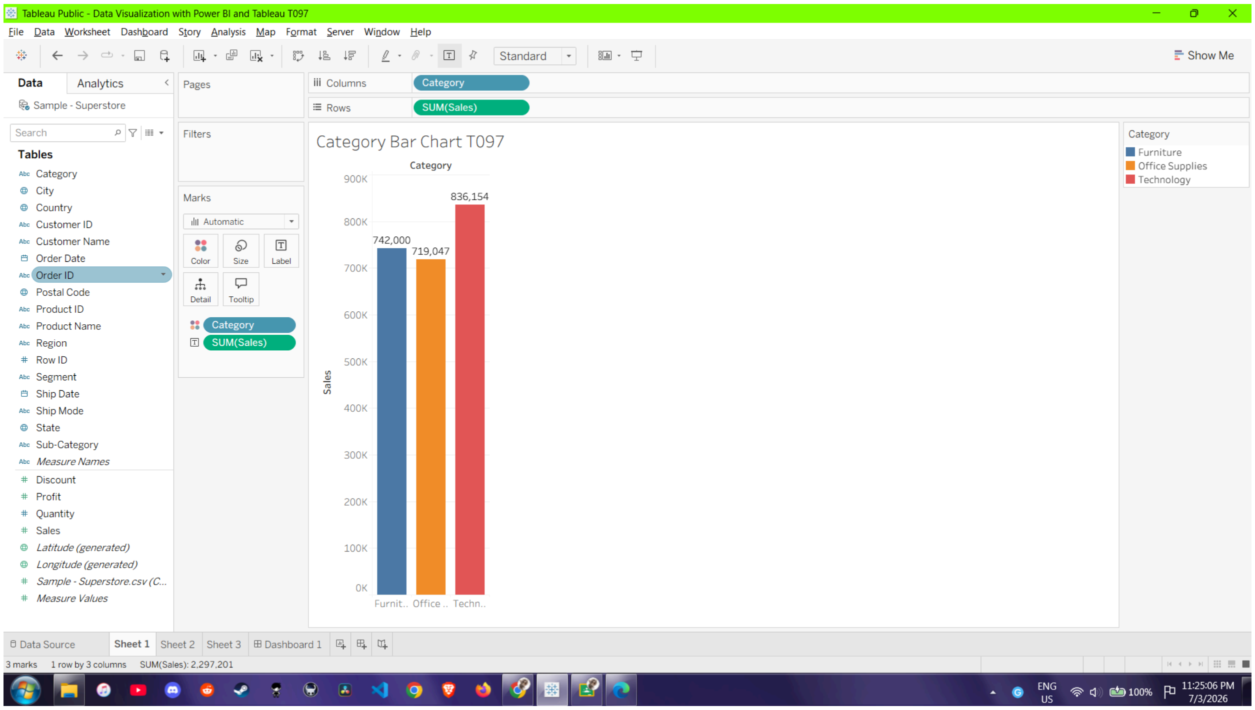
3. Convert fields between discrete and continuous and observe changes

Right-click the "Quantity" measure to toggle it between Discrete (blue blocks) and Continuous (green continuous axis) , dragging it to the Columns shelf each time to observe the visual difference.



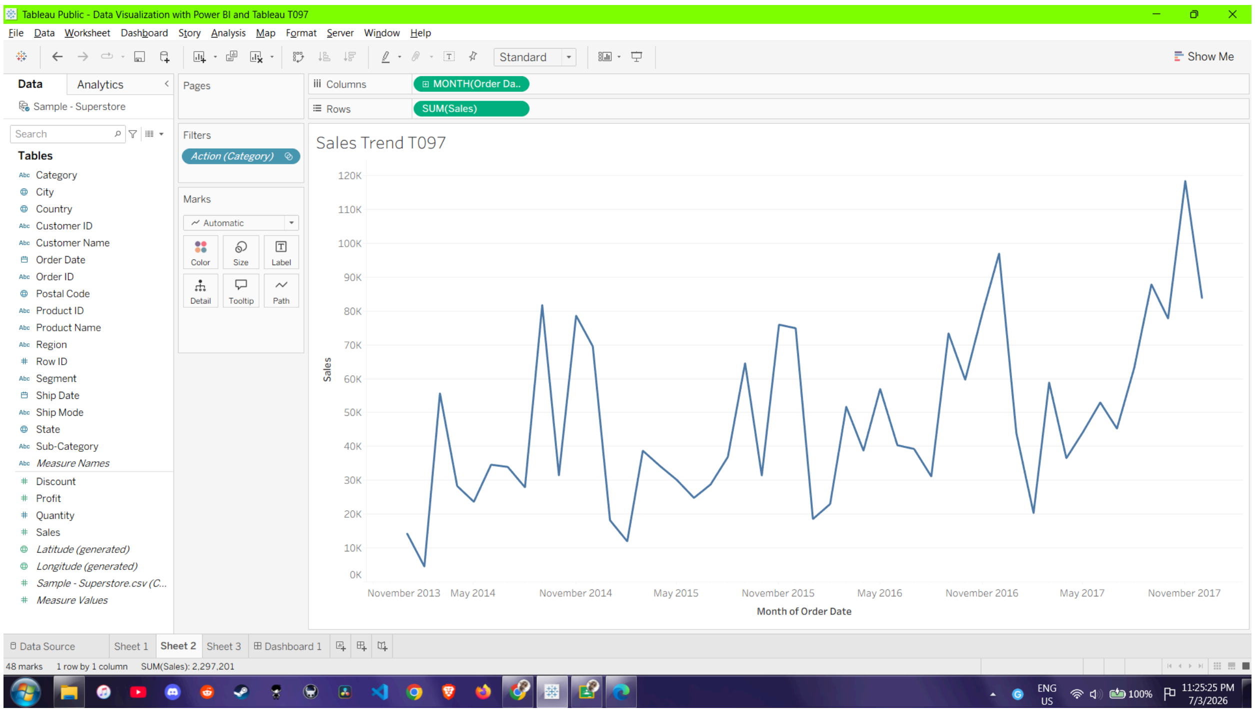
4. Create and customize a basic bar chart

Drag "Category" to Columns and "Sales" to Rows to generate a bar chart , then drop "Category" onto Color and "Sales" onto Label in the Marks card before renaming the sheet to "Category Bar Chart".



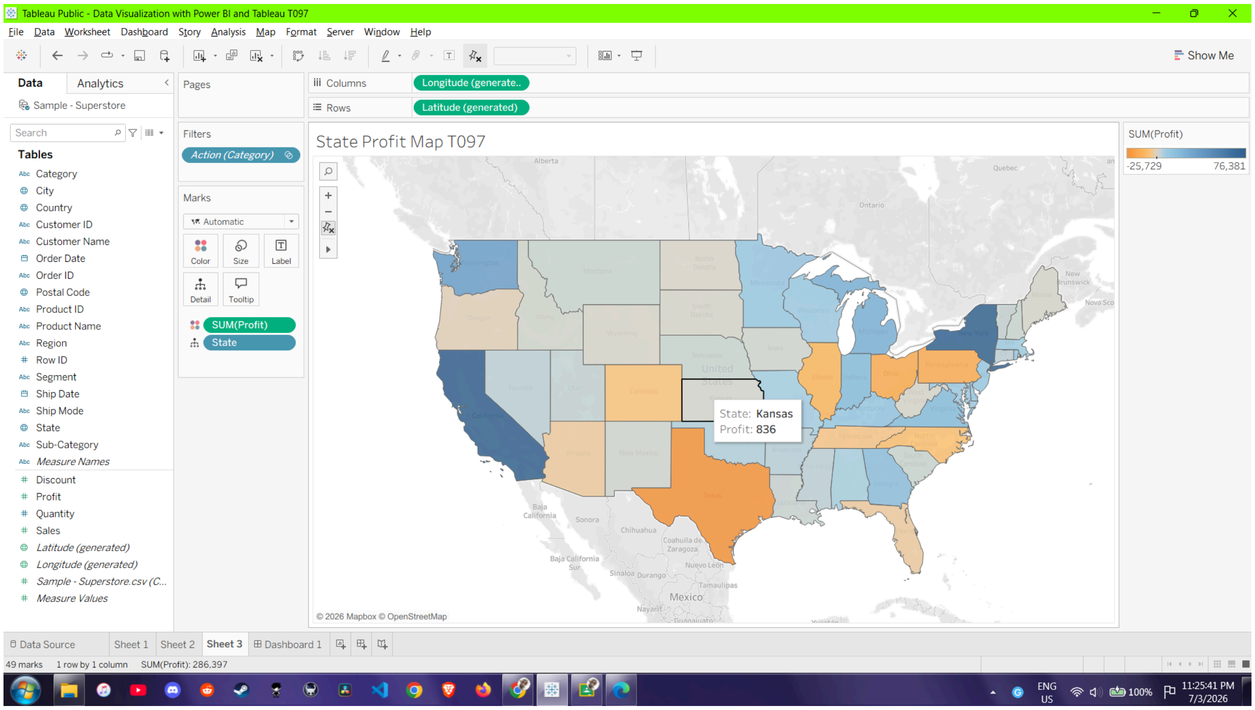
5. Build a line chart for time-series analysis

Open a new worksheet , drag "Order Date" to Columns and "Sales" to Rows to form a line chart, click the "+" icon on the date pill to drill down into quarters/months , and rename the sheet to "Sales Trend".



6. Create a geographic map visualization

On a new worksheet , double-click the "State" dimension to plot a map , drop "Profit" onto the Color tile to code the states , and rename the sheet to "State Profit Map".



7. Combine multiple charts into a simple interactive dashboard

Create a new Dashboard , drag and arrange all three of your completed sheets onto the canvas, and click the "Use as Filter" funnel icon on the bar chart so clicking a category dynamically updates the rest of the dashboard.